CONGRESSIONAL APP CHALLENGE 2026 · CA-10

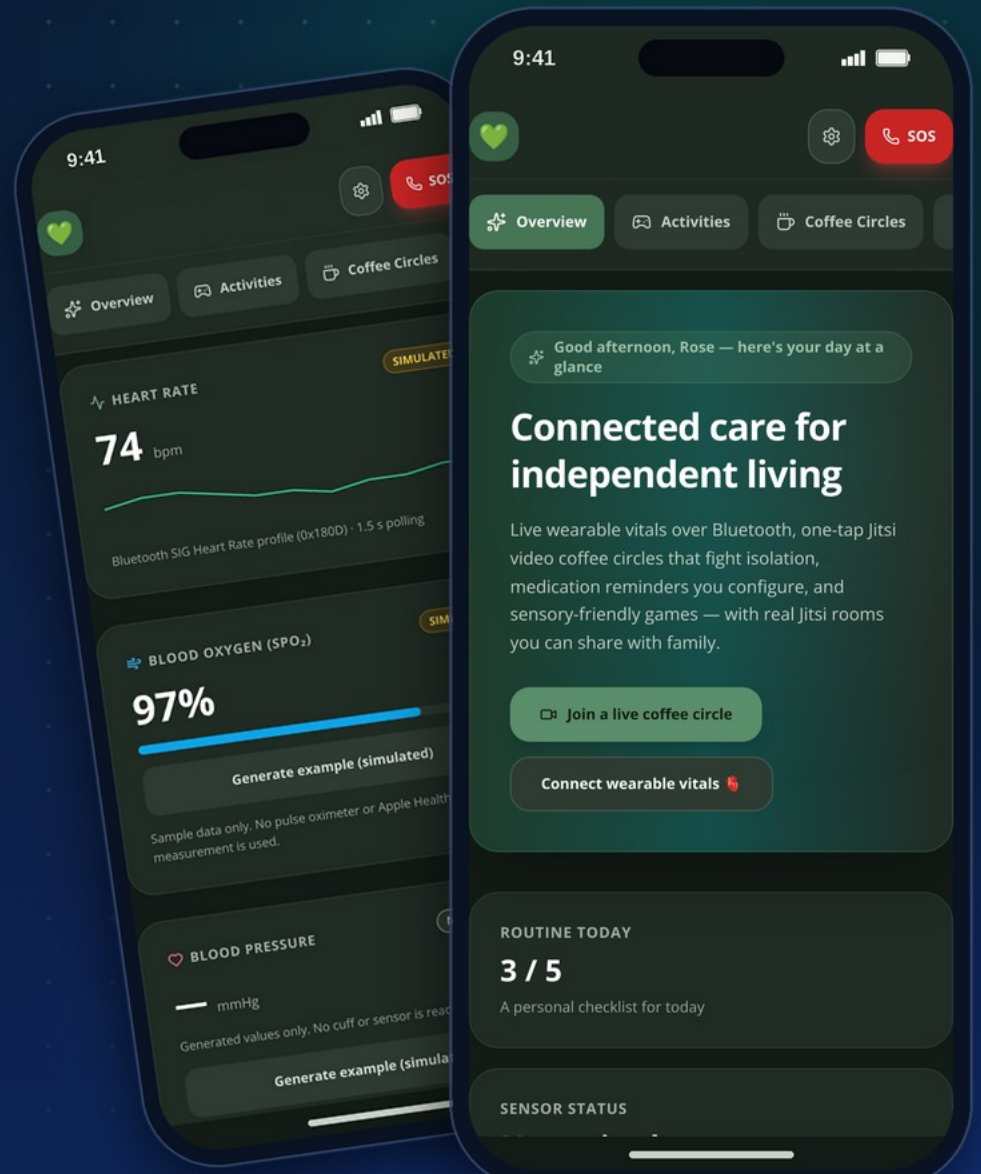
CareSphere AI

Connected care for independent living

A private care companion for older adults, autistic people and the families who care for them.

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THE DESIGN PROBLEM

02

Everyday support. Less friction.

Care should not feel like another task.

A predictable day

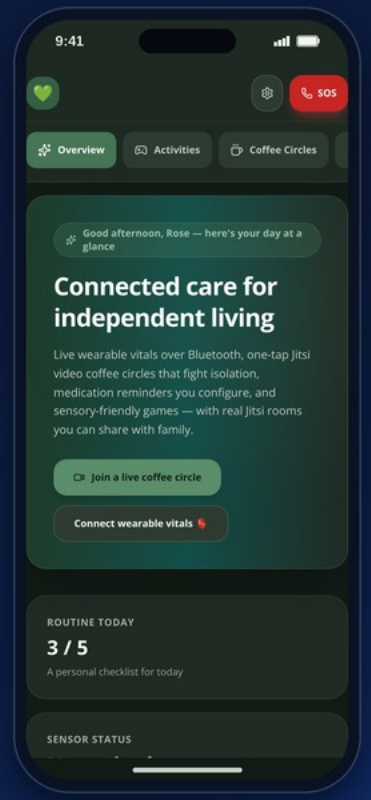
Mood check-ins, routines and medication self-check-ins.

A low-pressure activity

Self-paced games and sensory choices.

A way to connect

Video rooms and a brief you prepare before a visit.



WHAT THE APP DOES

03

Daily support. Play, connect, prepare.

01 **Activities & sensory tools**

Four self-paced games, breathing and optional soundscapes.

02 **Everyday routines**

Mood check-ins, daily tasks and medication self-check-ins.

03 **Human connection**

Coffee Circles open Jitsi video rooms; care notes stay local.

04 **Visit preparation**

Choose notes and readings for a previewable visit brief.

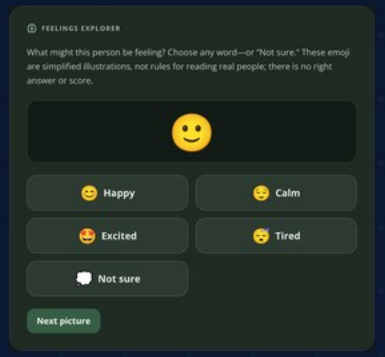
For older adults, autistic people and the families who support them.

CORE FEATURE · GAMES

04

Four ways to play. At your own pace.

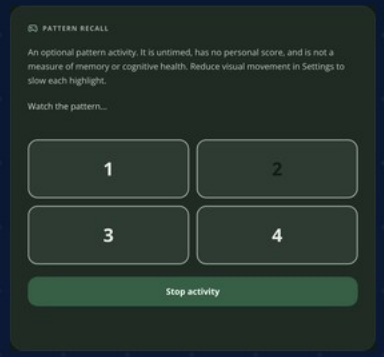
Feelings Explorer



Choose how you feel.

No right or wrong answer.

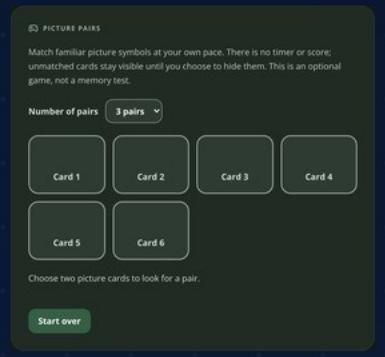
Pattern Recall



Watch, then repeat.

Replay or stop anytime.

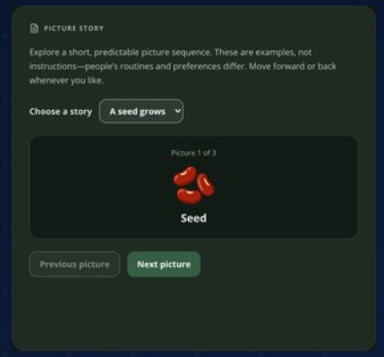
Picture Pairs



Reveal matching cards.

No countdown or score.

Picture Story



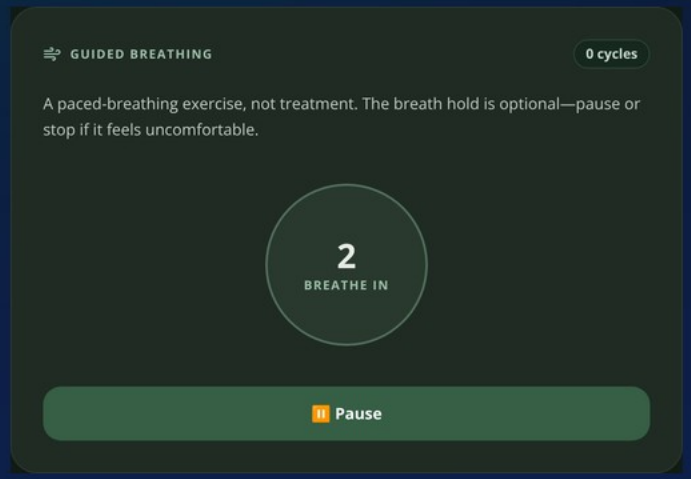
Explore a picture sequence.

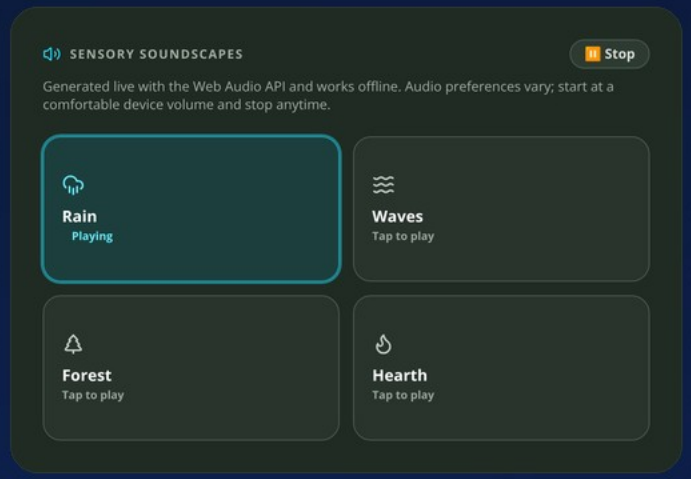
Advance when ready.

SENSORY TOOLS · RESEARCH-INFORMED DESIGN

05

A calmer experience. You choose.





Start. Pause. Stop. At your pace.

4–4–6 breathing · four optional soundscapes Text size, contrast, palette and motion choices.

Raymaker et al., 2019 · Alahmari et al., 2023 · Liu et al., 2025. Full references in notes and guide.

Research → design

AASPIRE: predictable navigation and sensory choice.

Game studies vary. Optional practice and enjoyment — not validated therapy.

LIVE APP DEMONSTRATION

Let me show you CareSphere in action.

Play → choose → check in → prepare

REAL APP · SAMPLE DATA

TOOLS, LANGUAGES AND PRIVACY

07

Two clients. Local-first care.

Web app

JavaScript

React · Vite · Web Audio Web Bluetooth · Web Crypto

Native iPhone app

Swift

SwiftUI · HealthKit · CoreBluetooth CryptoKit · optional local Health Guide

Encrypted local care data. Sharing is deliberate.

Jitsi calls and online references use external services. The screenshots shown are the web app.

PROGRAMMING CHALLENGE

08

Old connection. Wrong reading? Not here.

A late Bluetooth callback could update a newer session.

Reject stale sessions

A generation counter checks which connection owns the callback.

Keep the source honest

Every accepted sample is labeled LIVE_BLE or SIMULATED.

src/App.jsx · excerpt

```
if (generation !==
    hrConnectionGenerationRef.current)
  return
```

Current session only. Source labels remain in exports.

WHAT COMES NEXT

09

Build with people. Then evaluate.

Co-design	Invite intended users and caregivers to review the experience.
Usability testing	Observe whether tasks are understandable and comfortable.
iPhone beta	Prepare a TestFlight beta after iteration and device testing.

Lesson: a working feature is not proof of real-world benefit.



scan for source

Care that stays yours.

Connected, calm and local-first — with real readings or clearly labeled simulation.

EXPLORE THE PROJECT READ THE CODE BUILD LOCALLY

github.com/engineer2025-sudo/Care

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Explore the source, build instructions and project documentation.